



Elevating Learning Experiences: Leveraging Large Language Models as Student-Facing Assistants in Discussion Forums

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ABSTRACT

Recent advancements in instruction-tuned large language models offer new potential for enhancing students' experiences in large-scale classes. Deploying LLMs as student-facing assistants, however, presents challenges. Key issues include integrating class-specific content into responses and applying effective pedagogical techniques. This study addresses these challenges through retrieval and prompting techniques, focusing on mitigating hallucinations in LLM-generated responses, a crucial concern in education. Furthermore, practical deployment brings further challenges related to student data privacy and computational constraints. This research strives to enhance the quality and relevance of LLM responses while addressing practical deployment issues, with an emphasis on creating a versatile system for diverse domains and teaching styles.

KEYWORDS

Natural language processing, Educational tools, Discussion forum

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1 INTRODUCTION

In recent years, the enrollment growth in Computer Science (CS) in higher education institutions has outpaced teaching resources [1]. Most important is that the increase in the number of CS educators has not kept pace with student enrollment. Researchers have worked on building tools and examining different ways to increase students' sense of belonging in courses [6, 8]. Thus, our

work presents a Natural Language Processing (NLP) solution for assisting teaching staff in answering student questions in large-scale CS courses.

Although Large Language Models (LLM) demonstrate promising domain-specific question-answering capabilities [2, 4, 5], they have several obstacles to their broader application in education. The demands for answer accuracy in education are incredibly high, and hence, LLM hallucination (the tendency for LLMs to generate coherent, yet incorrect responses) is a significant challenge to producing high-quality answers to student questions [9]. One remedy is to fine-tune these models on large amounts of high-quality data of student questions answered by teaching staff. The problem here is that LLM fine-tuning (i) requires immense amounts of computing resources while the alternative of (ii) using publicly hosted LLMs might compromise sensitive student data and invade student privacy.

2 METHODS AND PRELIMINARY RESULTS

The framework for the student-facing and LLM-based discussion forum extension is presented in Figure 1. The framework includes two main modules - (i) the retrieval module for extracting relevant context for the student query and course materials along with (ii) the LLM prompting module to generate the response using an LLM. The framework was tested on a large-scale upper-division data science course (Principles and Techniques of Data Science) at the University of California, Berkeley. The enrollment in the course is around 1,200 students each semester.

2.1 Retrieval Module

First, we demonstrate that student forum Q&A data combined with course documents can be used to aggregate valuable context for assisting teaching staff in answering a student question. We develop a pipeline using non-LLM neural embedding methods to fetch documents and past student Q&A pairs similar to the new student question. This context would then be presented to the teaching staff to answer the question. We perform a thorough human evaluation of the pedagogical benefit of our novel use of unstructured student Q&A data.

Our human evaluation metric is inspired by similar evaluation criteria used in Insta-Reviewer [3]; the final evaluation criteria includes three metrics: *Relevance*, *Diversity*, and *Similarity to Ground Truth*. We evaluate the document retrieval alone and document

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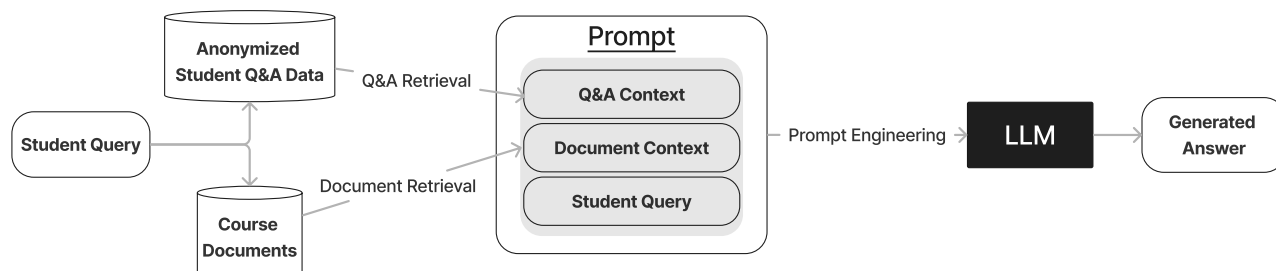


Figure 1: Relevant and broad context for the incoming student query is formed through Q&A and Document retrievals. The context is then prompted with the original student query to an LLM to generate an answer.

and Q&A retrievals combined to quantify the added benefit of unstructured Q&A data in providing relevant and diverse context to the student question used to generate a response similar to an expert. We observe improvements across all three metrics, with 68% of the retrieved context sufficient to replicate the ground truth response. The evaluations were conducted by four of the co-authors of the paper, who are members of teaching staff in Computing courses, and an inter-rater reliability protocol was applied before the start of the evaluations to ensure consistent scoring among the evaluators.

2.2 LLM Prompting Module

The second module leverages the retrieved context from the retrieval module to prompt the LLM to generate an answer. Similar to our retrieval module, we perform human evaluations to demonstrate the value of the retrieved context to the pedagogical quality of the generated response. We also include prompt engineering techniques using expert and subject pre-conditioning.

The evaluation criteria for this stage are also inspired by that of Insta-Reviewer [3]. Since, in this stage, we are evaluating the quality of a generated response rather than the context, we modified the evaluation rubric, and utilized *Relevance*, *Readability*, *Tone*, and *Factuality*. We observe improvements in the results of our framework compared to those of the no-retrieval (only prompt engineering) and zero-shot (only student question) techniques [7].

3 FUTURE WORK

Although we have a working framework that can fetch relevant Q&A pairs and course content and prompt an LLM to respond to student questions, we plan to continue our work to improve, extend, and deploy the framework. We observe that the retrieval component sometimes suffers from a lack of data; this issue can be mitigated by including Q&A data from previous offerings of the course, integrating more sources, including package documentations (for debugging questions) and relevant textbooks (for content questions), and employing more robust data indexing techniques. We believe that the prompting results can be improved by leveraging advanced prompting strategies and possibly fine-tuning the LLM in terms of factuality and employing pedagogical techniques to assist students. That said, we realize that fine-tuning language

models can be computationally costly and raise practical issues when setting up the system for a new class.

The framework can be released as a browser extension or as an EdSTEM bot. Before distributing the software for use in the classroom, the system would first be only accessible to the teaching staff. The purpose of this stage is to gather feedback from the course staff to identify any problems and improvements in the system. This would also allow course staff to review answers and make edits, which can subsequently be used for instruction tuning.

Additionally, we plan to evaluate the framework’s effectiveness in diverse educational settings, including project-oriented and math-intensive classes, to ensure its applicability to various learning environments.

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